House Price Prediction Using Machine Learning

Accurately predicting house prices is critical for buyers, sellers, and real estate professionals to navigate dynamic market conditions. This presentation explores advanced machine learning techniques to model property prices based on essential features. Our goal is to build robust, explainable models that support informed decision-making in real estate transactions while addressing challenges in data and model interpretation.

by - Savee Gupta

Introduction & Objective

Objective

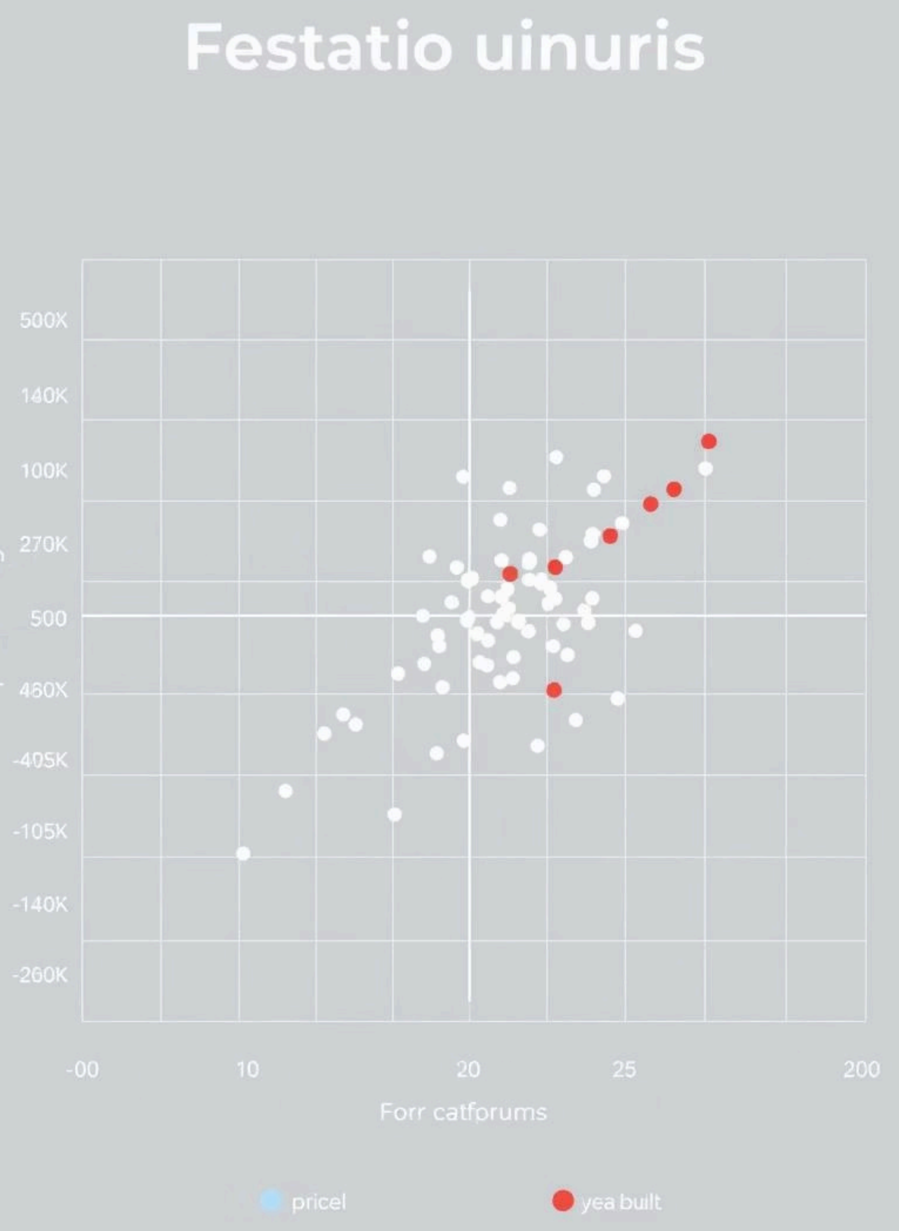
Develop a predictive model for property prices using features like size, bedroom count, bathroom count, neighborhood, and building type.

Motivation

Enable market participants to make data-driven pricing decisions amidst fluctuating market dynamics and varying property characteristics.

Approach

Leverage machine learning algorithms that provide accuracy and interpretability for practical application in real estate pricing.



Data Exploration & Key Insights

Dataset Features

- Size (square footage)
- Bedrooms and Bathrooms
- Neighborhood and Building Type
- Price (target variable)

Insights

- Size exhibits the strongest positive correlation with price
- Bedrooms, bathrooms, neighborhood, and building type also materially impact price

Visualizations

Correlation matrices and feature importance plots clarify variable relationships and guide model design.

Model Building Process

Data Preparation

Manage missing values, encode categorical variables such as neighborhood and building type for model readiness.

Model Selection

Use Linear Regression as a baseline to capture linear relationships; apply Random Forest for complex, nonlinear patterns.

Input & Output

- Input features: size, bedrooms, bathrooms, neighborhood, building
- Output: predicted property price

Model Evaluation & Limitations

Metrics

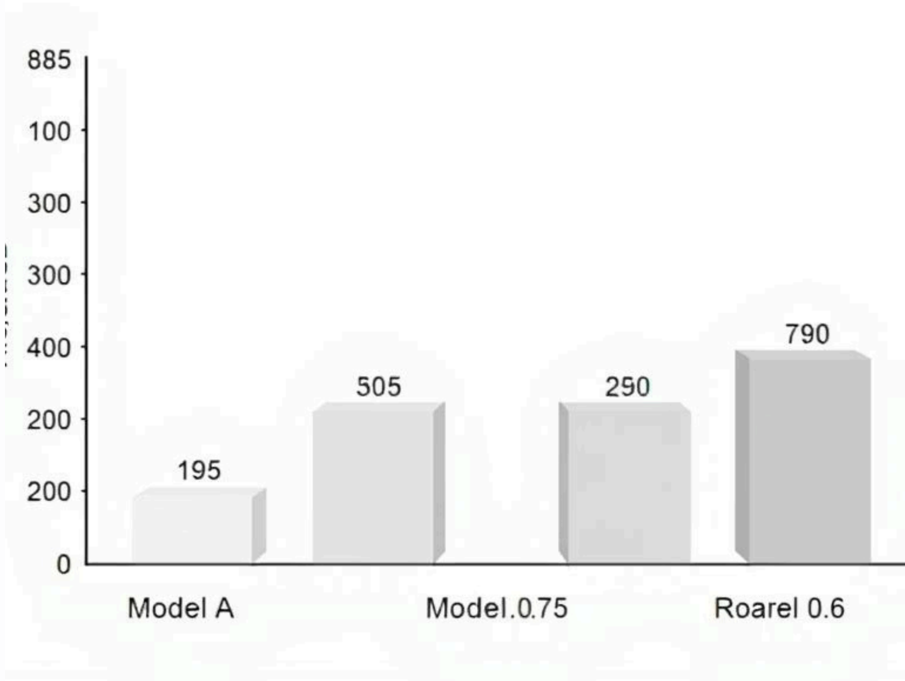
R^2 Score measures explained variance; RMSE gives average error in currency units for interpretability.

Findings

Random Forest consistently outperforms Linear Regression, capturing nonlinearities and feature interactions.

Shortcomings

Limited by feature set (no amenities, property age) and basic encoding; lacks cross-validation, reducing robustness.



Price Valuation Tool & Extensions

1

Purpose

Classifies properties by comparing actual and predicted prices using ±10% thresholds.

2

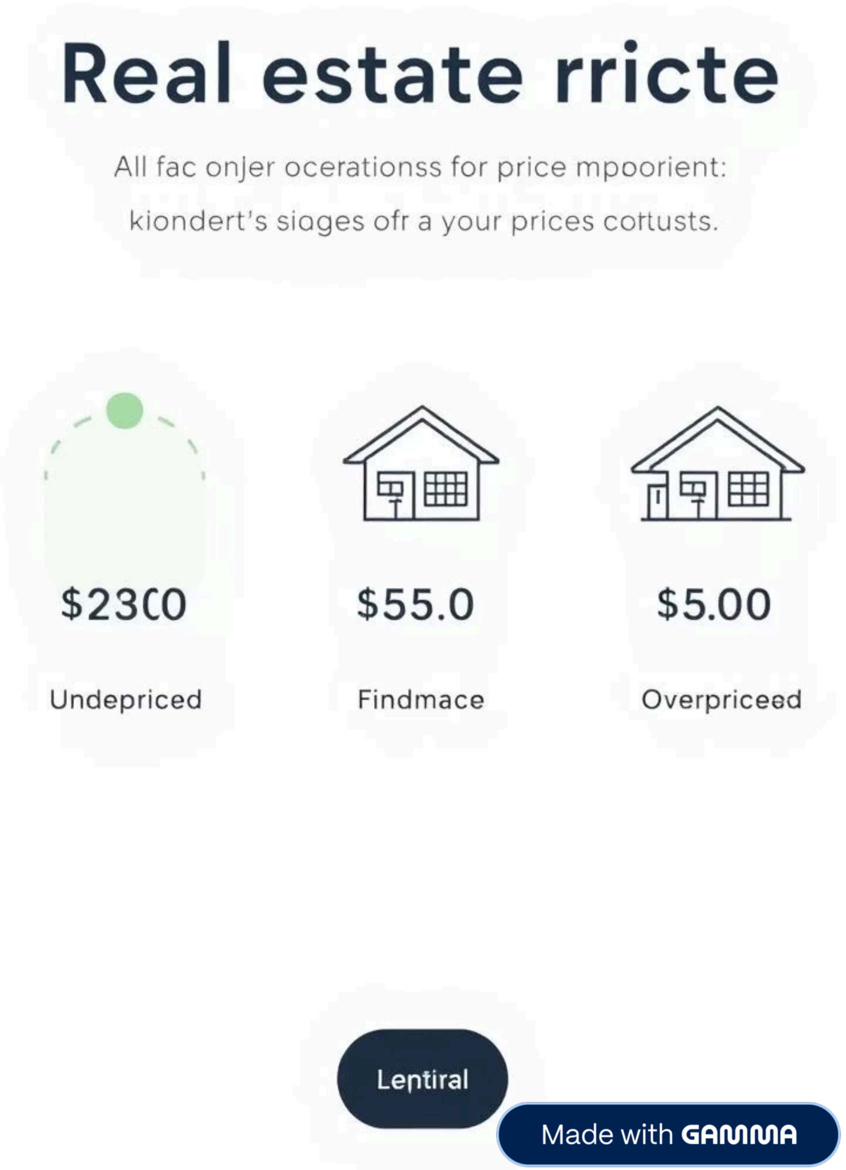
Functionality

Inputs property features, outputs price prediction and valuation label: Underpriced, Fairly Priced, or Overpriced.

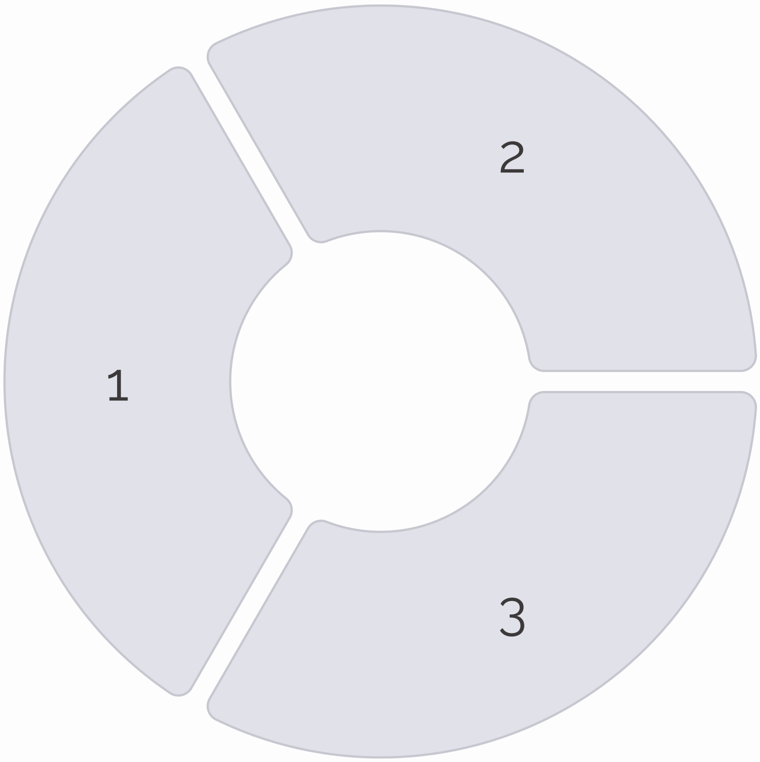
3

Future Extensions

Incorporate dynamic thresholds, market trends, and explainable AI for improved accuracy and transparency.



Explainability & Validation



Explainability

Use SHAP and LIME to attribute model decisions to key features, enhancing transparency for end users.

Validation Techniques

Employ holdout sets and cross-validation to ensure performance generalizes well beyond training data.

User Transparency

Provide confidence intervals, highlight top contributing features, and incorporate user feedback loops.

Deployment & System Architecture

System Flow

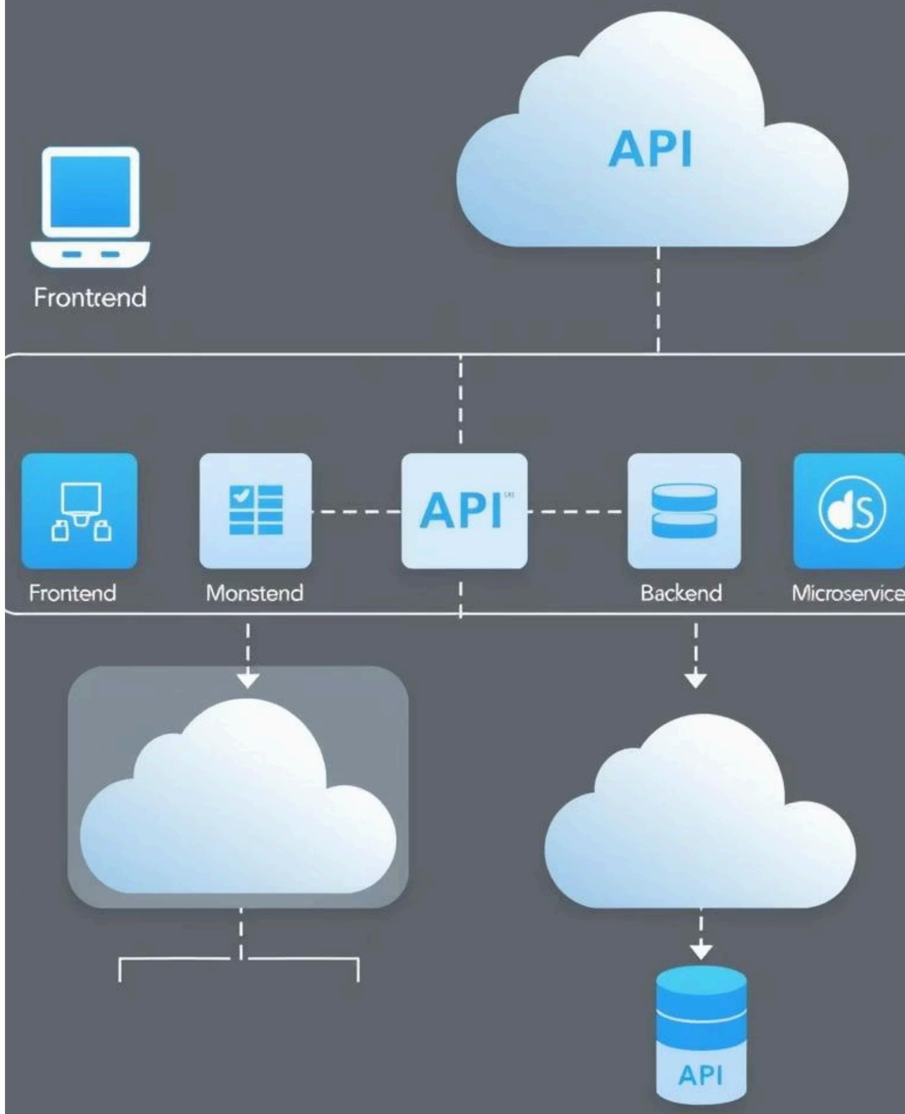
Web UI receives user inputs, Backend API handles prediction & valuation, Output presented with explanations.

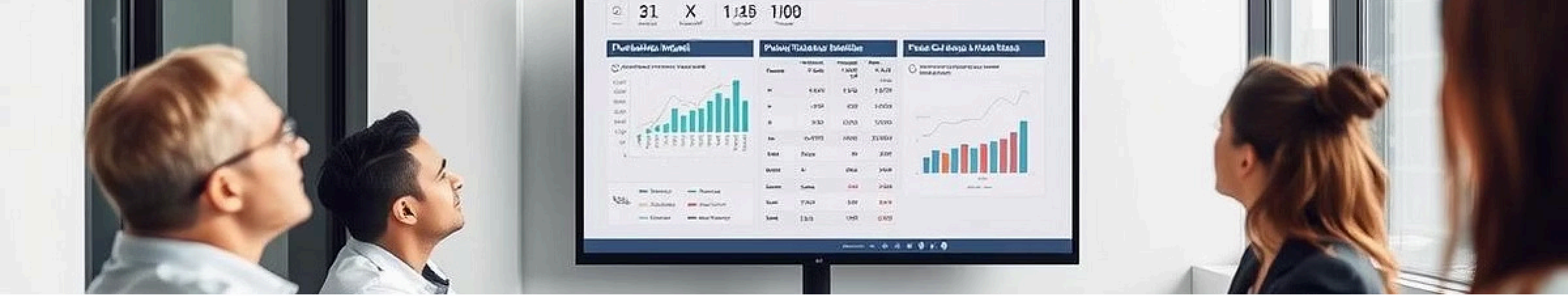
Scalability

Microservices architecture with load balancing and containerized cloud deployment ensures reliability and performance.

Monitoring & KPIs

Automated tracking via Prometheus and Grafana monitors prediction accuracy, uptime, latency, user engagement, and model drift.





Summary & Next Steps

Refine Feature Set

Incorporate additional variables like property age, amenities to boost model fidelity.

Enhance Model Robustness

Implement cross-validation and ensemble methods to improve accuracy and reliability.

Expand Deployment

Integrate market trend data, user feedback, and explainability features in production for real-world impact.

I look forward to addressing your questions and exploring how this predictive framework can transform real estate decision-making.